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Editorial

Diagnostic Guidelines for Periprosthetic Joint Infection: Know the Differences

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Guidelines play an important role in the diagnosis and treatment of periprosthetic joint infections (PJI). The diagnostic guidelines for PJI have undergone major evolution over the last three decades, becoming increasingly sophisticated and standardized [1–6]. The Musculoskeletal Infection Society in 2011 established one of the first standardized PJI criteria [2]. This widely used Musculoskeletal Infection Society definition was subsequently endorsed and modified during the first International Consensus Meeting (ICM) held in 2013 [3]. The PJI definition was further modified at the ICM in 2018 [5]. These modifications were performed based on an evidence-based validation study conducted before the meeting [7]. Interestingly, when looking at the orthopaedic literature, it appears that many authors were incorrectly interpreting the criteria mentioned in the validation study as the 2018 ICM PJI definition. This is evidenced by the disparity in the number of citations both articles have received in PubMed; the validation study garnered 1,808 citations [8], whereas the proceedings of the 2018 ICM have only 240 citations [9], as of the search conducted on January 20, 2025. Moreover, it is uncertain whether the authors are aware of the modifications made in the final 2018 ICM PJI definition (Figure 1).

A

Major criteria (at least one of the following)	Decision
Two positive cultures of the same organism	Infected
Sinus tract with evidence of communication to the joint or visualization of the prosthesis	

		Minor Criteria	Score	Decision
Preoperative Diagnosis	Serum	Elevated CRP *or* D-Dimer	2	≥6 Infected
		Elevated ESR	1	
	Synovial	Elevated synovial *WBC count or LE*	3	2-5 Possibly Infected [a]
		Positive alpha-defensin	3	
		Elevated synovial PMN (%)	2	0-1 Not Infected
		Elevated synovial CRP	1	

	Inconclusive pre-op score *or* dry tap [a]	Score	Decision
Intraoperative Diagnosis	Preoperative score	-	≥6 Infected
	Positive histology	3	4-5 Inconclusive [b]
	Positive purulence	3	
	Single positive culture	2	≤3 Not Infected

B

Major criteria (at least one of the following)	Decision
Two positive growth of the same organism using standard culture methods	Infected
Sinus tract with evidence of communication to the joint or visualization of the prosthesis	

Minor Criteria	Threshold		Score	Decision
	Acute[a]	Chronic		
Serum CRP (mg/L)	100	10	2	Combined preoperative and postoperative score: ≥6 Infected; 3-5 Inconclusive*; <3 Not Infected
or				
D-Dimer (ug/L)	Unknown	860		
Elevated Serum ESR (mm/hr)	No role	30	1	
Elevated Synovial WBC (cells/µL)	10,000	3,000	3	
or				
Leukocyte Esterase	++	++		
or				
Positive Alpha-defensin (signal/cutoff)	1.0	1.0		
Elevated Synovial PMN (%)	90	70	2	
Single Positive Culture			2	
Positive Histology			3	
Positive Intraoperative Purulence[a]			3	

Figure 1. (A) Initial scoring-based definition for periprosthetic joint infection proposed by Parvizi et al. in 2018. Reproduced from Parvizi et al. [7]. (B) The 2018 International Consensus Meeting criteria for periprosthetic joint infection. Reproduced from Shohat et al. [5]. CRP, C-reactive protein; ESR, erythrocyte sedimentation rate; WBC, white blood cell; PMN%, polymorphonuclear neutrophil percentage; LE, leukocyte esterase; AD, alpha-defensin.

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Table 1
A Comparison of the Validation Study Using the Scoring-Based Definition of PJI and the 2018 International Consensus Meeting Criteria for PJI.

Criteria	Parameters	Validation Study [7]				ICM 2018 [5]				Differences
Major criteria	*At least one of the following* 1. Two positive cultures of the same organism. 2. Sinus tract with evidence of communication to the joint or visualization of the prosthesis.				Infected				Infected	None
		Acute	Chronic	Score	Decision	Acute	Chronic	Score	Decision	
Minor criteria										No D-Dimer cutoff for acute cases in ICM 2018
Preoperative diagnosis									Combined preoperative and Postoperative score:	
Serum	Elevated CRP (mg/L) *or*	10	1,0	2		100	10	2		
	D-Dimer (ug/L)	860	860			-	860			
	Elevated ESR (mm/hour)	-	30	1	Preoperative score:	-	30	1		
Synovial fluid	Elevated synovial WBC (cells/μl)	10,000	3,000	3	≥6 Infected	10,000	3,000	3	≥6 Infected	1. WBC or LE or AD got only a single score
	Leukocyte esterase	++	++		2 to 5 Possibly infected	++	++		3 to 5	
	Positive alpha-defensin (signal-to-cut-off ratio)	1,0	1,0	3	0 to 1 Not infected	1,0	1,0		Inconclusive <3 Not infected	2. In chronic cases, PMN% cutoff changed from 80 to 70%
	Elevated synovial PMN%	90	80	2		90	70	2		
	Elevated synovial CRP (mg/L)	6,9	6,9	1						3. Synovial CRP is not included.
Intraoperative diagnosis	1. Single positive cultur			2	≥ 6 Infected			2		Note the final score changes for inconclusive and noninfected cases
	2. Positive histology			3	4 to 5 inconclusive			3		
	3. Positive purulence			3	≤ 3 not infected			3		

PJI, periprosthetic joint infections; CRP, C-reactive protein; ESR, erythrocyte sedimentation rate; WBC, white blood cell; PMN%, polymorphonuclear neutrophil percentage; LE, leukocyte esterase; AD, alpha-defensin; ICM, International Consensus Meeting.

Therefore, what are the differences between them? Major differences are in the preoperative diagnosis and decision based on the score (Table 1).

1. Preoperative diagnosis (serum): In the validation study, 860 (ug/L) was given as the D-Dimer threshold for acute cases, but this was removed from the final 2018 ICM definition.
2. Preoperative diagnosis (synovial fluid): Of the important differences, one is in alpha-defensin (AD). In the validation study, AD was a separate diagnostic test with a score of three points, but the ICM 2018 recommended using either elevated synovial white blood cell or leukocyte esterase or AD with three points.
3. Preoperative diagnosis (synovial fluid): In chronic cases, the ICM 2018 changed the threshold of polymorphonuclear neutrophil percentage from 80 to 70%.
4. Preoperative diagnosis (synovial fluid): Synovial C-reactive protein was included in the validation study, but this was excluded from the ICM 2018.
5. Combined preoperative and intraoperative score: In the validation study, a score of 4 to 5 and $\leq$ 3 were taken as "inconclusive" and "not infected," respectively, whereas the ICM 2018 recommends a score of 3 to 5 for inconclusive and < 3 as not infected.

Accurate infection diagnosis and the prevention of misdiagnosis constitute the primary objectives of these definitions. Moreover, it is essential to recognize that implementing the right definition in hip and knee arthroplasty research will assure uniformity across various studies and enhance the quality of existing evidence in the field. Future studies should consider this aspect, particularly when comparing the performance of the ICM 2018 with other PJI definitions [10]. Although the development of the ICM 2025 is ongoing, we strongly encourage all orthopaedic surgeons, infectious disease specialists, and researchers to be cognizant of this matter and make concerted efforts to use the appropriate definition for future studies.

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